

The derivative of a composite function

The chain rule — the one rule that makes everything else in calculus usable.

LESSON 10–12

3 × 40 MIN

ALGEBRA 11, §1.5

P1 · 7.5

LESSON CLOCK

14'	22'	26'	26'	30'	17'
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Why a new rule is needed 14 min

The chain rule 22 min

Practising it 26 min

Mixing with product and quotient 26 min

Related rates 30 min

Homework 17 min

LEARNING OBJECTIVES

- Identify the inner and outer function of a composite expression.
- Apply the chain rule in both notations.
- Combine the chain rule with the product and quotient rules.
- Use the chain rule for related rates of change.

TEXTBOOK

National pp. 55–70

Cambridge Pure Mathematics 1, pp. 146–155

Workbook P1 Exercise 7E

01 Explanation

Why the power rule is not enough

$y = (3x + 1)^5$ is a fifth power, so is it $5(3x + 1)^4$? Test the idea on the easiest case, $y = (3x)^2$. Expanding gives $9x^2$, whose derivative is $18x$ — but $2(3x) = 6x$. The answer is out by a factor of 3: exactly the derivative of the inside.

THE CHAIN RULE

$$(f(g(x)))' = f'(g(x)) \cdot g'(x) \text{ or } \frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$

Differentiate the outside, leaving the inside untouched — then multiply by the derivative of the inside.

Using it

Method. Name the inside u , write y in terms of u , differentiate both, multiply.

Y	U	$\frac{DY}{DU}$	$\frac{DU}{DX}$	$\frac{DY}{DX}$
$(3x + 1)^5$	$3x + 1$	$5u^4$	3	$15(3x + 1)^4$
$\sqrt{x^2 + 1}$	$x^2 + 1$	$\frac{1}{2\sqrt{u}}$	$2x$	$\frac{x}{\sqrt{x^2 + 1}}$
$\frac{1}{(2x - 5)^3}$	$2x - 5$	$-3u^{-4}$	2	$-\frac{6}{(2x - 5)^4}$
$(x^2 - 4x)^7$	$x^2 - 4x$	$7u^6$	$2x - 4$	$7(2x - 4)(x^2 - 4x)^6$

NEVER FORGET THE SECOND FACTOR

Omitting $\frac{du}{dx}$ is the single commonest error in the whole of A-level calculus. Every bracket, every root, every reciprocal needs it.

Mixing the rules

Real expressions need two rules at once. Decide the **outermost** structure first:

EXPRESSION	OUTERMOST STRUCTURE	RULES
$x^2(2x + 1)^5$	a product	product, then chain inside
$\frac{(x + 1)^3}{x^2}$	a quotient	quotient, then chain
$(x^2(x + 1))^4$	a power	chain, then product inside
$\sqrt{x(x + 3)}$	a root	chain, then product inside

$$(x^2(2x + 1)^5)' = 2x(2x + 1)^5 + x^2 \cdot 5(2x + 1)^4 \cdot 2 = 2x(2x + 1)^4(6x + 1)$$

ALWAYS FACTORISE THE ANSWER

The common bracket comes out, and the factorised form is what every later question — stationary points, sign of the gradient — actually needs.

Related rates

The Leibniz form makes the chain rule read as a genuine cancellation, which is why it handles rates that depend on other rates:

$$\frac{dV}{dt} = \frac{dV}{dr} \cdot \frac{dr}{dt}$$

Example. A balloon's radius grows at 2 cm/s. How fast is the volume growing when $r = 5$?

$$V = \frac{4}{3}\pi r^3 \Rightarrow \frac{dV}{dr} = 4\pi r^2 = 100\pi$$

$$\frac{dV}{dt} = 100\pi \times 2 = 200\pi \approx 628 \text{ cm}^3/\text{s}$$

SUBSTITUTE THE NUMBER LAST

Differentiate symbolically first, then put $r = 5$ in. Substituting first turns the variable into a constant and the derivative into zero.

02 Worked examples

EXAMPLE 1 Differentiate $y = (4x - 3)^6$.

$$\textcircled{1} \quad u = 4x - 3$$

$$\textcircled{2} \quad \frac{dy}{du} = 6u^5$$

$$\textcircled{3} \quad \frac{du}{dx} = 4$$

$$\textcircled{4} \quad 6(4x - 3)^5 \times 4$$

ANSWER

$$24(4x - 3)^5$$

EXAMPLE 2 Differentiate $y = x\sqrt{x^2 + 5}$.

① A product; the second factor needs the chain rule.

② $(\sqrt{x^2 + 5})' = \frac{x}{\sqrt{x^2 + 5}}$

③ $y' = \sqrt{x^2 + 5} + \frac{x^2}{\sqrt{x^2 + 5}}$

④ $= \frac{2x^2 + 5}{\sqrt{x^2 + 5}}$

Over a common denominator.

ANSWER

$$\frac{2x^2 + 5}{\sqrt{x^2 + 5}}$$

EXAMPLE 3 The side of a cube grows at 0.5 cm/s. How fast is the volume growing when the side is 8 cm?

① $V = a^3$

② $\frac{dV}{da} = 3a^2 = 192$

③ $\frac{dV}{dt} = 192 \times 0.5$

④ $= 96$

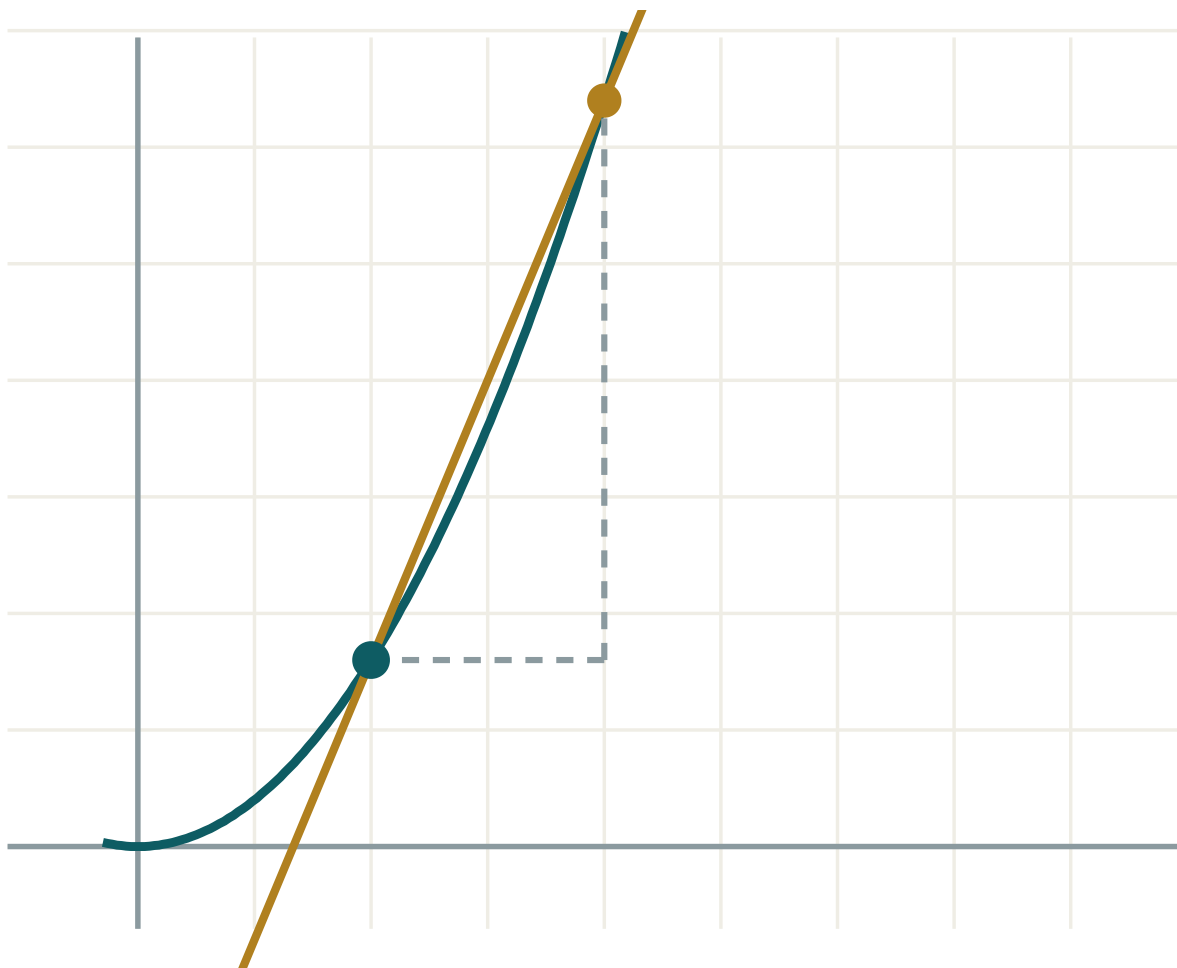
ANSWER

96 cm³/s

03 Interactive model

Differentiate one bracket by expanding and by the chain rule, and check the answers agree.

Composite gradients



x 2

h 2

 $\Delta y / \Delta x$ 2.4 $f'(x)$ exactly 1.6

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Chain rule	Zanjir qoidasi	Правило цепочки
Composite function	Murakkab funksiya	Сложная функция
Inner function	Ichki funksiya	Внутренняя функция
Outer function	Tashqi funksiya	Внешняя функция
Substitution u	Almashtirish	Замена переменной
Related rates	Bog'liq tezliklar	Связанные скорости
Bracket power	Qavs darajasi	Степень скобки
Intermediate variable	Oraliq o'zgaruvchi	Промежуточная переменная

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

$((2x + 1)^3)'$ is:

$$3(2x + 1)^2$$

$$6(2x + 1)^2$$

$$2(2x + 1)^2$$

$$3(2x + 1)^3$$

$(\sqrt{x^2 + 1})'$ is:

$$\frac{1}{2\sqrt{x^2 + 1}}$$

$$\frac{x}{\sqrt{x^2 + 1}}$$

$$\frac{2x}{\sqrt{x^2 + 1}}$$

$$\sqrt{2x}$$

The chain rule in Leibniz form is:

$$\frac{dy}{dx} = \frac{dy}{du} + \frac{du}{dx}$$

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$

$$\frac{dy}{dx} = \frac{du}{dy} \cdot \frac{dx}{du}$$

$$\frac{dy}{du}$$

$((x^2 + 3)^4)'$ is:

$$4(x^2 + 3)^3$$

$$8x(x^2 + 3)^3$$

$$2x(x^2 + 3)^3$$

$$4x(x^2 + 3)^3$$

For related rates you should substitute the number:

first

last

never

twice

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $((x + 1)^2)'$

ANSWER $2(x + 1)$

2. $((3x)^4)'$

ANSWER $324x^3$

3. $((2x + 5)^3)'$

ANSWER $6(2x + 5)^2$

4. $((x - 7)^5)'$

ANSWER $5(x - 7)^4$

5. $(\sqrt{2x})'$

ANSWER $\frac{1}{\sqrt{2x}}$

6. $((4 - x)^3)'$

ANSWER $-3(4 - x)^2$

7. $((x^2 + 1)^2)'$

ANSWER $4x(x^2 + 1)$

Medium · the standard question

7 PROBLEMS

1. $((3x - 2)^7)'$

ANSWER $21(3x - 2)^6$

2. $(\sqrt{x^2 + 9})'$

ANSWER $\frac{x}{\sqrt{x^2 + 9}}$

3. $(\frac{1}{(x + 2)^3})'$

ANSWER $-\frac{3}{(x + 2)^4}$

4. $((x^2 - 3x)^4)'$

ANSWER $4(2x - 3)(x^2 - 3x)^3$

5. $(x(x + 1)^3)'$

ANSWER $(x + 1)^2(4x + 1)$

6. $(\frac{(2x + 1)^2}{x})'$

ANSWER $\frac{(2x + 1)(2x - 1)}{x^2}$

7. Radius grows at 3 cm/s; find $\frac{dA}{dt}$ for a circle at $r = 4$

ANSWER $24\pi \text{ cm}^2/\text{s}$

Hard · stretch and reason

7 PROBLEMS

1. $(x^2\sqrt{x+1})'$

ANSWER $\frac{x(5x+4)}{2\sqrt{x+1}}$

2. $(\frac{\sqrt{x+1}}{x})'$

ANSWER $-\frac{x+2}{2x^2\sqrt{x+1}}$

3. $((x+1)^2+3)^4'$

ANSWER $8(x+1)((x+1)^2+3)^3$

4. $(\sqrt{x}\sqrt{x})'$

ANSWER $\frac{3}{4}x^{-1/4}$

5. Balloon: $\frac{dr}{dt} = 2$ cm/s. Find $\frac{dV}{dt}$ at $r = 10$

ANSWER 800π cm³/s

6. A ladder 5 m slides: foot moves at 0.4 m/s. Find the top's speed when the foot is 3 m out

ANSWER 0.3 m/s downwards

7. Show $(\frac{1}{g(x)})' = -\frac{g'(x)}{(g(x))^2}$

ANSWER Chain rule on g^{-1}

07 Homework**Homework — 6 tasks**

Factorise every answer. Task 6 needs the rate set out with its units.

1. Differentiate $y = (5x - 1)^4$.

2. Differentiate $y = \sqrt{3x^2 + 2}$.

3. Differentiate $y = \frac{1}{(x^2 + 1)^2}$.

4. Differentiate $y = x^3(2x - 1)^4$ and factorise the answer.

5. Differentiate $y = \frac{(x + 2)^3}{x}$.

6. A spherical balloon is inflated so that its radius grows at 1.5 cm/s. Find the rate of increase of its volume when $r = 6$ cm.